Checkpoint 2 — Maximum Likelihood Estimator for the Weights — Student Version

INFO 521 — Machine Learning Foundations · checkpoint (prompts & criteria; graded in class)

> **How this works**
>
> This checkpoint is sat **in class, closed-book** (~15–20 minutes, one derivation) and graded Satisfactory / Not-Yet against the criteria below. Version B is the alternate-version retake, offered the following week; you keep your best result. Knowing the prompt in advance is the point — practice until you can reproduce the derivation unaided.

Sitting: Week 3 · retake: Week 4 · gates Project 1.2.

Gates: Project 1.2 (Maximum Likelihood) **Format:** in-class, closed-book, ~15–20 min, one derivation. Graded Satisfactory / Not-Yet. **Retakeable.**

Prompt (Version A)

Assume $y_n = \mathbf{w}^\top \phi(x_n) + \epsilon_n$ with $\epsilon_n \sim \mathcal{N}(0, \sigma^2)$ i.i.d. Write the log-likelihood of the dataset and derive the maximum likelihood estimator $\hat{\mathbf{w}}$. State why it coincides with the least-squares solution.

Prompt (Version B — retake)

For the same model, derive the MLE for the noise variance $\hat{\sigma}^2$, treating $\hat{\mathbf{w}}$ as known.

Satisfactory criteria

- Correct dataset likelihood as a product over i.i.d. points; correct log.
- Maximizes (equivalently, drops constants and minimizes the negative log).
- Reaches the stated end result; (A) notes the σ^2 terms are constant in $\mathbf{w}$, so maximizing likelihood = minimizing squared error.